When Does Spectral Adaptation Help? Information Budgets and Mismatch Costs for Continuous Signal Representations

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Abstract

Spectral adaptation—tuning the frequency parameters of a continuous representation, allocating capacity per band, or injecting a spectral prior—gives large gains on some reconstruction tasks and almost none on others. Arbitrary-scale super-resolution (ASISR) is the clearest case of the latter: three years of decoder design on top of LIIF have moved the standard benchmark by roughly a third of a decibel, and a recent controlled study reports architectural saturation. We ask whether one can tell *in advance*, without training and without ground truth, whether spectral adaptation will help on a given task and frequency band, how much it can help, and what it costs when the spectral prior is wrong. We (i) show that the common forms of spectral adaptation amount to choosing a prior spectrum (Bochner); (ii) compute a per-band linear recoverability R_b^{lin} from the observation operator alone, which upper-bounds every stationary-kernel linear estimator; and (iii) derive a closed-form mismatch cost $\Delta(R, \kappa)$ whose asymmetry splits bands into three regimes: wide/low-bias representations are safe where R is high, the spectrum must be matched in between, and only conservative priors or learned (non-linear) priors are safe where R is low. Pre-registered experiments on single-image ASISR at $\times 2$–$\times 8$, sparse-pixel reconstruction, multi-frame super-resolution and 1D EEG confirm the predicted mismatch-cost shape and asymmetry, locate the decoder bottleneck of ASISR in the low-R regime where four decoders (LIIF, LTE, SC-INR and a variant) differ by at most 0.03 in band recovery, and locate the benefit of the cell/footprint readout (LTE and SC-INR, seed 1) in the bands that the trained network, not the linear budget, partially recovers. We report the pre-registered criteria that failed alongside those that held.

1 Introduction

Continuous (implicit) representations decode a signal at arbitrary coordinates, and a large body of work controls *which frequencies* such a representation can express: the frequency scale of sinusoidal activations [Sitzmann et al., 2020, Liu et al., 2024], the bandwidth of Fourier features [Tancik et al., 2020], target-spectrum-driven hyperparameter selection [Kania et al., 2025, TODO, 2025, 2026a], per-band capacity allocation [TODO], or explicit spectral priors such as Gaussian processes and kriging [Rasmussen and Williams, 2006]. We refer to all of these as *spectral adaptation*. Its empirical record is uneven. In multi-frame super-resolution, choosing the Fourier-feature bandwidth well is decisive [TODO, 2026b]; in sparse-pixel reconstruction, a matched spectrum is worth several decibels (§5.3); but in single-image arbitrary-scale super-resolution (ASISR) the same lever barely moves.

ASISR is the setting this paper grew out of. Starting from LIIF [Chen et al., 2021], a line of decoders—LTE's dominant-frequency estimator [Lee and Jin, 2022], our scale-consistent SC-INR with an analytic footprint (sinc) readout, and rotation-equivariant decoders (Rot-E) [TODO: cite Rot-E]—each injected some form of spectral or geometric structure into the decoder. On the shared EDSR-baseline/DIV2K protocol, three years of such designs moved Urban100 $\times 4$ by about 0.35 dB (MetaSR 26.05 dB [Hu et al., 2019] vs. CiaoSR 26.58 dB [Cao et al., 2023], as reported by the original papers) [src: LIT-B] and a controlled re-training study finds the best method only 0.035 dB ahead of the second [Nasir et al., 2026]. Our own controlled arms reproduce the picture at the level of frequency bands: at $\times 4$, four decoders recover the band one octave above the LR Nyquist limit to within 0.02 of each other [src: OUT7.1/h6c]. The natural question is *why*: is the decoder the wrong place to adapt the spectrum, or is there nothing left to adapt?

We argue that the answer is determined by an information budget that can be computed before any training. For a known linear observation $y = Ax + n$, the fraction of each band that *any* stationary-kernel linear estimator can recover is bounded by the LMMSE recoverability R_b^{lin} under a generic $1/f^2$ prior. This number depends only on A, the noise level and the generic prior, and is cheap to compute. Combined with a closed-form cost of using a wrong spectrum, it yields a three-regime map that says, per task and per band, whether spectral adaptation can help, by how much at most, and in which direction an error is dangerous.

Contributions.

1. **A training-free diagnostic.** We cast spectral adaptation as the choice of a prior spectrum (Prop. 1) and use the per-band linear recoverability R_b^{lin} (Prop. 2) to draw an a-priori three-regime map for single-image, sparse-pixel, multi-frame and 1D signal tasks (§4, Fig. 1).

2. **A closed-form mismatch cost.** Prop. 3 gives the excess error of a prior that is off by a factor κ in a band of recoverability R. Its asymmetry yields a per-band tendency—*wide* is safe as $R \to 1$, *matching* matters at intermediate R, *conservative* is safe where R is low—and its energy-weighted sum over bands ranks representations (though it does not reliably pick the single best); it also explains why grid fields collapse under noise and why large-σ Fourier features fail (§4).

3. **Pre-registered cross-task validation** (§5). The mismatch-cost shape and asymmetry hold on 60 task–band–κ cells (Spearman 0.90) [src: E2]; the zones move with the ASISR scale factor as predicted (Spearman 0.952 over 27 cells) [src: E3]; in the low-R regime trained ASISR networks exceed the linear oracle in every cell while differing among themselves by at most 0.03 in 10 of 12 cells [src: E3], which accounts for decoder saturation; and the benefit of the cell/footprint readout is zero in the high-R band and appears only where the network has partially recovered the band—consistently in LTE and SC-INR (cross-architecture Spearman 0.985, seed 1) [src: F1]; averaging the full model over training scales predicts the ranking of these losses after the fact (Spearman 0.82; correlational) [src: L1L2].

Scope and honesty. Props. 2–3 follow directly from classical Wiener/LMMSE theory; our claim is not new estimation theory but its use as an a-priori diagnostic and prescription for spectral adaptation in continuous representations, together with a pre-registered test across tasks. Several pre-registered criteria failed; we report them in the main text and in §6.

2 Related Work

Spectral control in implicit representations. SIREN [Sitzmann et al., 2020] sets the frequency scale ω_0 by hand; Fourier features [Tancik et al., 2020] expose the bandwidth σ and connect it to the neural tangent kernel [Jacot et al., 2018]; FINER [Liu et al., 2024] widens the reachable frequency set. A recent group selects these parameters from the spectrum of the target signal: FreSh [Kania et al., 2025] matches the spectrum of the untrained model to the target's, WINNER [TODO, 2025] scales weight noise by the spectral centroid, SEC [TODO, 2026a] maps the centroid to the optimal parameter via a calibration set. WIEN-INR [TODO] allocates capacity per band. All of these address *fitting* a fully observed signal. We study the inverse setting, where the spectrum is to be estimated from degraded observations and the question is whether it can pay off at all. Spectral bias [Rahaman et al., 2019] explains what an untuned network prefers; our budget explains what the observation permits.

Arbitrary-scale super-resolution. LIIF [Chen et al., 2021] decodes a local latent at continuous coordinates with a cell (footprint) input; LTE [Lee and Jin, 2022] adds a local dominant frequency estimator; MetaSR [Hu et al., 2019], SRNO [Wei and Zhang, 2023], CiaoSR [Cao et al., 2023] and HIIF [Jiang et al., 2025] refine the decoder; SC-INR (ours, prior work [TODO: cite]) replaces the cell input with an analytic footprint (sinc) readout and enforces scale consistency; Rot-E [TODO: cite] adds rotation equivariance. Gaussian-splatting decoders [Chen et al., 2025] and heat-field decoders with analytic anti-aliasing [Becker et al., 2025] form a newer family. A controlled re-training study [Nasir et al., 2026] reports architectural saturation of the MLP/operator family. Low-pass behaviour of SR networks has been measured with a global scalar [TODO, 2024]; we measure recovery per band defined by the sampling theorem and set it against a computable linear bound. Our analysis explains saturation rather than proposing another decoder.

Multi-frame and sparse reconstruction. SuperF [TODO, 2026b] fits a Fourier-feature network to multiple shifted low-resolution frames and tunes σ; Deep Image Prior [Ulyanov et al., 2018] reconstructs without training data; kriging/GP regression [Rasmussen and Williams, 2006] inserts the spectrum directly as a prior. These are the high- and mid-R regimes of our map.

Classical theory and per-band diagnostics. LMMSE/Wiener estimation and aliasing analysis of sampling are textbook [Kay, 1993]. Per-frequency agreement measures such as Fourier ring correlation in microscopy [van Heel and Schatz, 2005] and scale-decomposed skill scores in forecast verification are close relatives of our per-band recoverability; to our knowledge none has been used to predict when spectral adaptation of continuous representations helps.

3 Spectral Adaptation as Choosing a Prior

Setting. A signal x lives on a continuous domain $\Omega \subset \mathbb{R}^d$. We observe $y = Ax + n$ with a known linear operator A (blur and decimation, sparse sampling, or the footprint integrals of several shifted frames) and $n \sim \mathcal{N}(0, \sigma_n^2 I)$. A continuous representation f_θ is fitted by minimising

$\|Af_\theta - y\|^2$ (possibly with regularisation or early stopping) and queried anywhere.

Proposition 1 (Fourier features are a GP prior; Bochner)**.** *Let* $f(u) = \sum_{j=1}^{m} w_j \varphi_j(u)$ *with* $\varphi_j(u) = \sqrt{2/m}\cos(2\pi b_j^\top u + c_j)$, $b_j \sim p(b)$, $c_j \sim \mathcal{U}[0, 2\pi)$ *and* $w \sim \mathcal{N}(0, s^2 I)$. *As* $m \to \infty$, f *is a zero-mean stationary Gaussian process with covariance* $k(h) = s^2\, \mathbb{E}_b[\cos(2\pi b^\top h)]$ *and spectral density* $S(\nu) = s^2 p(\nu)$.

Hence a ridge read-out of Fourier features is the GP posterior mean under prior spectrum S; with $p = \mathcal{N}(0, \sigma^2 I)$, $S(\nu) \propto \exp(-\|\nu\|^2/2\sigma^2)$. Tuning σ (SuperF) or choosing frequency parameters from a target spectrum (FreSh, WINNER, SEC) thus picks a prior spectrum within a restricted family; a GP with a fitted or known spectrum picks it directly. For an MLP on Fourier features trained by gradient descent, the lazy-training limit gives kernel regression with a stationary NTK modulated by p [Tancik et al., 2020, Jacot et al., 2018], so the statement holds to first order; we measure departures from it empirically (§5.4).

Proposition 2 (Linear upper bound)**.** *If* x *is a stationary Gaussian process with spectrum* P, *every linear estimator* $\hat{x} = Gy$ *satisfies* $\mathbb{E}\|x - \hat{x}\|^2 \geq \mathbb{E}\|x - \hat{x}_{\mathrm{LMMSE}}(P)\|^2$, *with* $\hat{x}_{\mathrm{LMMSE}}(P) = C_P A^\top (A C_P A^\top + \sigma_n^2 I)^{-1} y$.

For the class of Prop. 1 the true-spectrum LMMSE is therefore a ceiling, and the gap between two prior spectra is the cost of spectral mismatch.

Definition 1 (Per-band linear recoverability)**.** For a band projector Π_b,

$$R_b^{\mathrm{lin}}(A, \sigma_n, P) = 1 - \frac{\mathbb{E}\|\Pi_b(x - \hat{x}_{\mathrm{LMMSE}}(P))\|^2}{\mathbb{E}\|\Pi_b x\|^2}.$$

R_b^{lin} depends only on A, σ_n and P. With the generic natural-image prior $P \propto 1/f^2$ it requires no ground truth and no training; we compute it by sampling from the prior, simulating the observation and applying the LMMSE, in seconds per task. When a trained or otherwise nonlinear estimator is evaluated on real images, we report the same quantity measured on data, $R_b = 1 - \|\Pi_b(x - \hat{x})\|^2/\|\Pi_b(x - \bar{x})\|^2$ on the luminance channel; for a trained network we write R_b^{net}.

Bands. Throughout we use sharp radial sub-bands in HR pixel units, $b_1 = [0, \frac{1}{8})$, $b_2 = [\frac{1}{8}, \frac{1}{4})$, $b_3 = [\frac{1}{4}, \frac{1}{2}]$ cycles/pixel. At $\times 4$, b_1 lies below the LR Nyquist limit and b_2, b_3 lie one and two octaves above it.

Closed form for decimation. For $A =$ blur K followed by decimation by s, each LR frequency ω collects s^d HR frequencies $\nu_m = \omega + m/s$, and

$$R(\nu_0) = \frac{|\hat{K}(\nu_0)|^2 P(\nu_0)}{\sum_m |\hat{K}(\nu_m)|^2 P(\nu_m) + \sigma_n^2}. \tag{1}$$

The sum in the denominator is aliasing. In single-image and sparse-pixel settings it dominates the sensor noise, which is why R_b^{lin} there is nearly insensitive to σ_n (Table 1).

4 Mismatch Cost and Three Regimes

Consider one frequency (or a band treated as a scalar channel) with signal variance p and effective noise q—sensor noise plus aliased energy—so that $R = p/(p+q)$. An estimator that assumes variance κp instead of p applies the gain $g = \kappa R/(\kappa R + 1 - R)$.

Proposition 3 (Mismatch cost)**.** *The excess mean-squared error, normalised by the band energy* p, *is*

$$\Delta(R, \kappa) = \frac{R(1-R)^2(\kappa - 1)^2}{(1 - R + \kappa R)^2}. \tag{2}$$

A proof is in Appendix A. Two limits carry the prescription:

- under-estimation, $\kappa \to 0$: $\Delta \to R$; harmless when R is low, costly when R is high;
- over-estimation, $\kappa \to \infty$: $\Delta \to (1-R)^2/R$; harmless when R is high, and larger than the band energy itself when $R < 0.38$, i.e. the reconstruction is worse than not reconstructing the band at all.

In absolute units, with $Q = \kappa P$ the assumed and P the true energy and N the effective noise at one frequency,

$$P\,\Delta = \frac{N^2(Q - P)^2}{(P + N)(Q + N)^2}. \tag{3}$$

In the over-estimation window $P < N < Q$ the excess error is $\approx N$, independent of the signal energy; in the under-estimation window $Q < N < P$ it is $\approx P$. Over-estimation is paid in units of noise, under-estimation in units of signal [src: DER§1.2].

Corollary 1 (Where the asymmetry flips)**.** *In the per-frequency scalar model, for every* $\kappa > 1$, $\Delta(R, \kappa) > \Delta(R, 1/\kappa)$ *if and only if* $R < 1/2$.

The proof follows from the sign of $(\kappa - 1)(1 - 2R)$ (Appendix A). The statement is exact under two conditions: it is evaluated per frequency, and the mismatch is a symmetric multiplicative factor (κ vs. $1/\kappa$). Averaged over a band in which R varies, Jensen's inequality shifts the crossover, so band-level comparisons near $R = 1/2$ are only approximate (§5.1). The model also assumes that the effective noise q does not depend on the assumed spectrum; random sparse sampling, which aliases energy across bands, violates this, and the one counter-example to the band-level asymmetry in E2, sparse 10% b_1, lies there (§5.1).

Three regimes. We fix two thresholds before the experiments, 0.8 and 0.1 (sensitivity in §6):

- **Regime A** ($R_b^{\text{lin}} \geq 0.8$): the information is in the observation. In the limit $R \to 1$ over-estimating the spectrum costs little and under-estimating it costs much, which favours wide, low-bias representations; the bottleneck is then capacity and optimisation. This is a per-band tendency, not a rule for choosing a representation: at $R = 0.8$ a $16\times$ over-estimate is not free, and the winner is decided by the energy-weighted cost summed over *all* bands, each with its own κ_b (Eq. (4)).

- **Regime B** ($0.1 \leq R_b^{\text{lin}} < 0.8$): both directions cost; the gain of spectral adaptation is bounded by LMMSE(true) − LMMSE(mismatched), and matching the spectrum decides the outcome.

- **Regime C** ($R_b^{\text{lin}} < 0.1$): a linear estimator obtains almost nothing. Under-estimation is free, over-estimation is catastrophic; only priors that exploit cross-band statistical dependence—learned, non-linear—can recover the band.

Zone labels do not pick winners. To compare representations with effective priors Q we use the band-summed, absolute cost

$$C(Q) = \sum_b E_b \, \Delta\big(R_b, \kappa_b(Q)\big), \tag{4}$$

with band energy E_b, matched-prior recoverability R_b and $\kappa_b(Q)$ the representation's band energy relative to E_b; lower C is predicted better. For linear arms the exact trace formula $\mathbb{E}\|x-\hat{x}\|^2 = \operatorname{tr}\big((I-H_Q A)C_P(I-H_Q A)^\top\big) + \sigma_n^2\|H_Q\|_F^2$ needs no approximation. Eq. (4) ranks representations (E2: 0.90; D11: 0.856) but does not reliably select the best among close candidates (D11: 1/3), does not cover non-linear iterative fits, and its a-priori version (generic spectrum for E_b, κ_b) has not been tested (§5.4) [src: DER§2.3]. The per-frequency form Eq. (3) explains why over-estimation is paid in units of noise but is not used as evidence: for non-diagonal operators (random shifts, sparse sampling) its per-frequency predictions are inaccurate.

Three known failures as over-estimation. (i) Grid or hash fields without spectral decay assume a flat spectrum, i.e. $\kappa \gg 1$ in the high bands, and collapse once noise lowers R there; (ii) Fourier features with large σ in SuperF-style fitting do the same; (iii) a regulariser of fixed strength corresponds to a fixed κ relative to a noise-dependent p/q, so its effective mismatch drifts with noise. §5.4 tests these on multi-frame SR.

Fig. 1: three-regime map (Table 1 as a heat map, four task families including EEG). [TODO: render from d8 JSON]

Figure 1: A-priori regime map.

Two ways to assign a regime. R_b^{lin} describes what the observation permits a linear estimator; it is the right axis for linear, multi-frame, sparse and 1D tasks, and it is computable in advance. A trained ASISR network, however, lifts regime-C bands to intermediate recovery through its learned prior (§5.2). To describe what components *inside* such a network do, the regime has to be assigned by the network's actual recovery R_b^{net}, which requires ground truth and is therefore a post-hoc description, not a prediction (§5.5). At this network layer Prop. 3 is not a theorem: a network is not a linear Wiener estimator under some prior, and has no well-defined κ. What carries over is a boundary statement only: a component cannot matter in bands with $R_b^{\text{net}} \approx 1$ (the output does not depend on prior details) or $R_b^{\text{net}} \approx 0$ (nothing is recovered to lose), so ablation losses can appear only at intermediate R_b^{net}. No quantitative ranking and no direction of asymmetry follow [src: DER§3.2].

5 Experiments

Protocol. Every confirmatory experiment was pre-registered: the criteria, thresholds and analysis scripts were frozen (with hashes) before results were produced, and are not changed after the fact. Images are from DIV2K validation (0801–0812) unless stated; R_b is measured on luminance. Exploratory analyses are labelled as such.

5.1 The mismatch cost is predictable (E2)

Setup. 12 DIV2K images, five tasks (single image $\times 2$, $\times 4$; sparse pixels 10%, 25%, 50%). In each of the three bands we multiply the true spectrum by $\kappa \in \{1/16, 1/4, 4, 16\}$, reconstruct with a GP (linear), and measure the lost recoverability in that band, giving 60 cells [src: E2§2]. Band energies are computed after a Hann window on both signal and error; the window was not specified in the pre-registration and was added after a one-image run showed boundary leakage (single-image $\times 4$ b_3 measured $R = 0.56$ at 0.004% transmittance). After the fix, single-image $\times 4$ b_2 measures $R = 0.085$ against an a-priori 0.08 [src: E2§1]. Only the metric changed, not hypotheses or criteria.

Table 1: A-priori three-regime map: R_b^{lin} per band under the generic $1/f^2$ prior, no data, no training [src: OUT3/d8]. Shading: A ≥ 0.8, B, C < 0.1.

Task	σ_n	b_1	b_2	b_3
Single image ×2	.002 / .01	1.00 / 1.00	.92 / .90	.09 / .08
Single image ×4	.002 / .01	.98 / .98	.09 / .08	.01 / .01
Single image ×8	.002 / .01	.79 / .77	.01 / .01	.01 / .01
Sparse 5%	.002	.70	.09	.03
Sparse 10%	.002	.81	.20	.07
Sparse 25%	.002	.92	.47	.20
Sparse 50%	.002	.98	.76	.46
Multi-frame ×4, T=4	.002→.02	≈1.0	.73→.51	.12→.03
Multi-frame ×4, T=8	.002→.02	≈1.0	.83→.62	.27→.08
Multi-frame ×4, T=16	.002→.02	≈1.0	.90→.73	.41→.15

Table 2: E2 pre-registered criteria (60 cells, 12-image means) [src: E2§2; E4-T1].

Criterion	Threshold	Result
a. Spearman(band-avg. prediction, measured)	≥ 0.8	**0.90**
b. $R < 0.3$: over-est. > under-est.	$\geq 80\%$	4/4
b. $R > 0.7$: under-est. > over-est.	$\geq 80\%$	6/7
c. median |pred.−meas.|	≤ 0.05	0.038
d. Spearman(a-priori prediction, measured)	≥ 0.7	0.79

Table 3: Measured / band-averaged predicted cost Δ (fraction of band energy) [src: E2§3].

Task, band	R	κ=$\frac{1}{16}$	$\frac{1}{4}$	4	16
SI×2 b_2	.80	.230/.402	.047/.101	.067/.031	.165/.056
SI×2 b_3	.11	.084/.097	.044/.058	.194/.380	**1.370**/2.648
SI×4 b_2	.085	.060/.074	.029/.045	.150/.336	**0.933**/2.590
Sparse 10% b_2	.37	.197/.291	.072/.144	.110/.302	.586/.976
Sparse 25% b_2	.70	.226/.430	.068/.139	.094/.074	.473/.149
Sparse 50% b_2	.90	.148/.264	.036/.046	.024/.009	.064/.015

Results. All four pre-registered criteria hold (Table 2). In low-R bands under-estimation is nearly free (≤ 0.06) while a 16× over-estimate costs 0.9–1.4 times the band energy, i.e. worse than not reconstructing the band (Table 3). In high-R bands the direction reverses. When $R \approx 0$ (single image ×4 b_3) neither direction costs anything, because the observation carries nothing about the band [src: E2§3]. The band-averaged prediction over-estimates large-κ costs; for decimation, evaluating Eq. (2) per frequency with Eq. (1) and energy-weighting removes this Jensen gap (exploratory: Spearman 0.99, median error 0.004; e.g. SI×4 b_2, $\kappa = 4$: predicted 0.151, measured 0.150) [src: E2§4]. The single counter-example to the asymmetry is sparse 10% b_1 ($R = 0.92$; under 0.072 vs. over 0.110): random sampling spreads aliasing across bands, outside the scalar model [src: E2§2].

Table 4: Single image ×4, R_b measured on data [src: OUT7.1/h6c]. Regime by R_b^{lin}: b_2, b_3 are C. gen./orac.: LMMSE with generic/oracle spectrum; SC-eq/SC-ns: SC-INR-eq / SC-INR-nosinc.

Set	band	gen.	orac.	LIIF	LTE	SC-eq	SC-ns
Set14	b_2	.16	.25	.56	.57	.57	.56
	b_3	.09	.09	.20	.20	.20	.18
BSD100	b_2	.13	.22	.38	.38	.38	.38
	b_3	.06	.06	.13	.13	.13	.11
Urban100	b_2	.11	.27	.50	.50	.51	.51
	b_3	.02	.03	.14	.14	.15	.13

5.2 Regime C: why ASISR decoders saturate (M2, E3)

Setup. We compare four trained decoders that share the EDSR-baseline encoder and the LIIF training protocol (DIV2K, 1000 epochs, scale range ×1–×4, seed 1): LIIF, LTE, SC-INR-eq and SC-INR without the sinc footprint readout [TODO: confirm arm descriptions] against two linear references, LMMSE with the generic spectrum and LMMSE with the image's own (oracle) spectrum, on 256^2 centre crops of the first 30 images of Set14, BSD100 and Urban100 [src: OUT7.1/h6c].

×4. In both regime-C bands the trained networks recover far more than the oracle linear bound: 0.38–0.57 vs. 0.22–0.27 in b_2 (1.7–2.3×) and 0.13–0.20 vs. 0.03–0.09 in b_3 (2–6×) (Table 4). This recovery cannot come from spectral adaptation of a linear read-out (Prop. 2); it comes from the learned prior. The four decoders differ by at most 0.02 in either band.

Across scale factors (E3). We repeat the measurement at ×2, ×3, ×8 (27 cells: 3 scales × 3 bands × 3 datasets) with the same networks; ×8 is out of the training range [src: E3]. Pre-registered results (Table 5): the a-priori R_b^{lin} ranks the measured generic-LMMSE recovery with Spearman 0.952; in 10/12 regime-C cells all

Table 5: E3 pre-registered criteria [src: E3; E4-T1].

Criterion	Threshold	Result
a. Spearman(R_b^{lin}, generic LMMSE)	≥ 0.8	0.952
b. C: nets > oracle, range ≤ 0.03	$\geq 80\%$ cells	10/12
c. A: best net − oracle ≤ 0.05	$\geq 80\%$ cells	9/9
d. generic LMMSE monotone in scale	9/9	9/9
e. (expl.) oracle−generic: B>A, B>C	–	.078 vs .011/.040

four networks exceed the oracle LMMSE and differ by at most 0.03; in 9/9 regime-A cells the best network is within 0.05 of the oracle LMMSE (largest gap +0.016); generic-LMMSE recovery decreases monotonically with scale in 9/9 band–dataset pairs. The zones move as predicted: b_2 is A at $\times 2$ ($R_b^{\text{lin}} = 0.92$), B at $\times 3$ (0.42) and C at $\times 8$ (0.011); b_1 enters B at $\times 8$ (0.79) [src: E3-key]. In all 12 C cells every network exceeds the oracle (by +0.002 to +0.244) [src: E4-T2].

Scale collapse (L1, pre-registered). If regime C is set by the observation rather than by the decoder, recovery should depend on frequency relative to the LR Nyquist limit, $u = 2s\nu$, not on absolute frequency. On 240^2 crops of the first 30 images per dataset, we measure $R(u)$ in seven u bins at $\times 2/\times 3/\times 4$ for LIIF, LTE and SC-INR (seed 1) and take the median range across scales, D_{in} [src: L1L2]. The linear control collapses (D_{in} 0.019 / 0.006 / 0.011 on Set14 / BSD100 / Urban100). Binning by u instead of ν shrinks the range by an order of magnitude in 9/9 model–dataset groups (0.013–0.061 vs. 0.31–0.46). The main criterion, $D_{\text{in}} \leq 0.05$ in $\geq 8/9$ groups, fails: 6/9, with all three models on Set14 at 0.055–0.061 (BSD100 and Urban100: 0.013–0.019). This exceeds both the seed noise floor (median range across seeds 0.003–0.004) and the linear control; the cause is not diagnosed. We found no consistent deviation outside the training range ($\times 6/8/12$; $D_{\text{out}} > D_{\text{in}}$ in only 6/9), and the public models generalise only partly (GSASR and Thera pass, ContinuousSR fails; 6/9) [src: L1L2]. Above the LR Nyquist limit, our three decoders exceed the generic linear estimate by +0.18 to +0.40 and are almost indistinguishable from each other [src: L1L2].

Reading. In regime C the room for any decoder-side spectral adaptation is set by what the learned prior already recovers, and the four decoders sit within 0.03 of each other there. This is the band-level counterpart of the 0.035 dB saturation reported by Nasir et al. [2026]. The "within 0.03" statement depends on the threshold: at 0.02 only 8/12 cells pass [src: E4-E3sens]; see §6. Note also that regime A is not where networks pull ahead either: there the oracle linear estimator is already within 0.016 of the best network.

Table 6: Sparse-pixel reconstruction: R_{b_2}, R_{b_3} and PSNR (dB) for generic / true-spectrum linear / DIP [src: OUT7.2].

Rate	b_2	b_3	PSNR
5%	.17/.39/.16	.10/.20/.05	24.59/26.39/25.19
10%	.29/.56/.34	.14/.31/.12	26.13/28.45/27.02
25%	.59/.81/.69	.29/.57/.38	29.27/32.66/30.84
50%	.85/.95/.89	.57/.82/.65	33.37/37.89/34.79

5.3 Regime B: matching the spectrum decides (M3, M4)

Sparse pixels. Sampling rates 5–50%, DIV2K 0801–0812, luminance; we compare linear estimation with the generic spectrum, with the image's own spectrum, and DIP [Ulyanov et al., 2018] [src: OUT7.2/sp_scan]. The true-spectrum linear estimator beats DIP at every rate (26.39 vs. 25.19, 28.45 vs. 27.02, 32.66 vs. 30.84, 37.89 vs. 34.79 dB at 5/10/25/50%), and the generic spectrum trails it by 1.8–4.5 dB (Table 6). The space between the two linear estimators is the whole headroom of spectrum estimation in this regime.

1D EEG. On 200 calibration segments of EEGM-MIDB [Goldberger et al., 2000, Schalk et al., 2004] with 10/25/50% of samples kept, the spectral prior helps most in θ/α at 10–25% and almost nowhere at 50%, where most bands are already A; e.g. α at 10%: linear interpolation .22, population-average spectrum .30, true spectrum .47 [src: OUT7.3]. [TODO: add H8 test-set (760 segments) numbers from H8 result note; full table to appendix.]

5.4 Regime A↔B: multi-frame SR (E1)

Setup. DIV2K 0801–0812, $\times 4$, $T = 16$ shifted frames synthesised neutrally (footprints integrated on the 512^2 original, ground truth by 2×2 averaging), known registration, RGB, PSNR with an 8-pixel border crop; $\sigma_n \in \{0, 0.005, 0.01, 0.02\}$ [src: E1]. Arms: a multi-resolution grid field (mrgrid); least squares with TV (ls_tv); a SuperF-style Fourier-feature MLP (ff_σ); GPs with Gaussian spectra (gp_gauss_σ), power-law spectra (gp_power_α) and the image's own spectrum (gp_oracle); bilinear. Paired bootstrap 95% CIs over images.

Results. The direction predicted by Prop. 3 holds: representations without spectral decay (mrgrid, weak TV) collapse under noise, losing 17 dB between $\sigma_n = 0$ and 0.02 (Table 7), the over-estimation failure; the value of the true spectrum over the best power-law GP grows with noise, D from 0.33 to 0.85 dB, difference +0.51 [0.13, 0.88] (P3 holds); and the winner moves from a low-bias arm (gp_gauss_30) to gp_oracle (P4 holds) [src: E1§1]. At $\sigma_n = 0$ a low-bias GP beats the best Fourier-feature MLP by +2.76 [1.99, 3.56] dB.

Table 7: E1: 12-image mean PSNR (dB), $T = 16$ [src: E1§2, e1_result_v2.json].

Arm	σ_n=0	.005	.01	.02
mrgrid	31.39	28.36	22.57	14.36
ls_tv1e-4	31.16	27.37	21.24	13.78
ls_tv1e-3	29.98	29.61	28.64	24.84
ff_3	27.94	27.92	27.88	27.74
ff_10	28.86	28.79	28.61	28.01
gp_gauss_30	**31.62**	27.73	26.65	25.56
gp_power_2.0	31.19	29.37	27.37	26.10
gp_power_2.5	29.62	29.36	28.65	27.45
gp_power_3.0	28.27	28.21	28.04	27.50
gp_oracle	29.91	**29.70**	**29.25**	**28.35**
bilinear	24.65	24.64	24.59	24.41

Two pre-registered predictions fail [src: E1§1]. (P1) At $\sigma_n = 0.005$ low-bias representations no longer win: mrgrid − ff_10 = −0.43 [−1.01, 0.14] dB; the switch happens earlier than predicted. (P2) At $\sigma_n = 0.02$ the generic power-law GP does not come within 0.6 dB of the oracle (−0.85 [−1.10, −0.61]). The winner margins at $\sigma_n = 0$ and 0.005 are not significant (+0.19 and +0.10 dB, CIs include 0).

Why the switch came early. The pre-registered prediction used one zone label: at $T = 16$, $R^{\text{lin}}_{b_2}$ falls from 0.90 to 0.73 over the noise range (Table 1), so b_2 is regime A at small noise and low-bias arms were expected to win. It ignored each arm's mismatch κ_b and ignored b_3. The D11 diagnostic below shows that both matter. Two further notes. P4's switch at 0.005 partly reflects gp_oracle's numerical noise floor at $\sigma_n = 0$ (it is optimal among linear arms for $\sigma_n > 0$ by Prop. 2), so future switch predictions should compare practical arms only [src: DER§2.2]. For the non-linear arms, mrgrid and ls_tv1e-4 already lose more at $\sigma_n = 0.005$ than their noiseless lead over ff_10, which is the direct numerical reason P1 fails; their mechanism was not tested [src: DER§2.2].

Diagnostic on E1 (D11, pre-registered). We stored per-frequency error spectra of the GP arms on the E1 images and settings (CPU; the non-linear arms mrgrid, ls_tv and ff were not measured) [src: D11]. At $\sigma_n = 0.005$ the excess error of gp_gauss_30 over gp_oracle splits 0.5% / 44% / 56% over $b_1/b_2/b_3$. In b_3 the measured oracle recovery is $R = 0.22$ (regime B) with over-estimation $\kappa = 4.5$; in b_2, $R = 0.80$ but $\kappa = 16.4$ (Table 8). The pre-registered rule looked only at b_2; b_3 was already in regime B, and in b_2 a large κ is not free even at $R = 0.8$, since the "harmless over-estimation" of regime A is the $R \to 1$ limit. Pre-registered results: (a) the b_3 share of gp_gauss_30's excess is 0.556 [0.465, 0.619], above 0.5 at the point estimate but with a CI that includes it; (b) the band-wise predicted excess $\sum_b E_b\, \Delta(R_b, \kappa_b)$ ranks the measured excess over 24 arm–noise points with Spearman 0.856 [0.744, 0.918] (threshold 0.8, holds); (c) it does *not* pick the best non-oracle GP arm (1/3 noise levels, threshold 2/3): the top arms are close and the band-averaged form carries a Jensen bias (e.g. gp_power_2.0 at $\sigma_n = 0.005$: predicted 1.40×10^6, measured 3.9×10^4, unnormalised) [src: D11]. The concentration of excess error on $R(\nu) < 1/2$ frequencies (0.82 / 0.71 for the two wide arms vs. 0.18 for gp_power_3.0) has no discriminative value, because 78% [74%, 81%] of all frequencies satisfy $R(\nu) < 1/2$ at this noise level; we do not use it as evidence. So the band-wise cost $C(Q)$ (Eq. (4)) ranks the GP arms but does not select the winner; whether its a-priori version ranks equally well has not been tested [src: DER§2.3].

Non-linear arms (mrgrid, ls_tv, ff) are covered only qualitatively; $T = 4/8$ (4 images) is descriptive only.

Table 8: D11: gp_gauss_30 vs. gp_oracle, E1 multi-frame, $T = 16$, $\sigma_n = 0.005$, 12-image mean [src: D11].

Band	oracle R_b	κ_b	share of excess
b_1	1.00	1.02	0.5%
b_2	0.80	16.4	44%
b_3	0.22	4.5	56%

5.5 Inside the network: SC-INR's footprint readout (F1)

SC-INR's only component with a reliable benefit in our earlier ablations is its footprint readout (analytic sinc integration over the target pixel); in LTE the analogous input is the cell. F1 asks where in the band map this component matters. We ablate it in two architecture pairs (LTE vs. LTE-NoCell; SC-INR vs. SC-INR-NoSinc) and measure the per-band loss $L = R_{\text{full}} - R_{\text{ablate}}$ on 27 confirmatory cells ($\times 2/\times 3/\times 4 \times 3$ bands $\times$ 3 datasets), seed 1 [src: F1]. The pre-registered predictor is $g(R) = R(1-R)^2$, the small-mismatch limit of Eq. (2).

Results (LTE pair). L is zero where $R \approx 1$: mean L over regime-A cells is 0.0000, versus 0.0050 at $\times 3$ b_2 (regime B) [src: F1; E4-T1]. The loss concentrates in b_3 at $\times 2$–$\times 4$ and b_2 at $\times 4$, where R^{lin}_b is only 0.01–0.10 (regime C) but the network recovers 0.13–0.56. The a-priori predictor fails: Spearman$(g(R^{\text{lin}}_b), L) = 0.411 < 0.5$ (standard Spearman with average ranks for ties). Using the network's own recovery raises it to 0.490, still below the threshold [src: E4-T1].

Results (SC-INR pair). With the matched pair (SC-INR vs. SC-INR-NoSinc, same encoder), the picture is the same: the A-regime criterion holds, the a-priori predictor fails (Spearman$(g(R^{\text{lin}}_b), L) = 0.400$), the network-recovery predictor also stays below threshold (0.492), and the per-cell losses of the two architectures agree closely

(Spearman 0.985) [src: F1-corr2]. [TODO: A-regime mean L and $\times 3$ b_2 L for the matched SC pair: not in the corrected note.] All F1 numbers are seed 1 only; seeds 2/3 are evaluated under the G1 pre-registration below.

A closed-form band-order predictor (G1, pending). A training-free quantity, the relative gain difference between a one-pixel box footprint and point sampling, $\epsilon(\nu) = 1 - \mathrm{sinc}(\nu_x)\,\mathrm{sinc}(\nu_y)$, grows with frequency and orders the bands. On seed 1, after the fact, its band average ϵ_b^2 ranks the losses better than $g(R_b^{\mathrm{lin}})$ or $g(R_b^{\mathrm{net}})$ (Spearman 0.664 for LTE) [src: G1 prereg §0]; this is post hoc and not counted. A confirmatory test on seeds 2/3 (G1) was frozen beforehand. We use ϵ_b^2 only as a predictor of band order, not as a mechanism: a linearised gain-mismatch model of the missing readout, with a factor R or $R(1-2R)^2$, was tested and rejected (pre-registered L2b: Spearman 0.62–0.65 vs. 0.82 for the direct measurement below), and across all closed-form quantities we tried, multiplying by any R factor gives no consistent gain (0.60–0.71 vs. 0.65–0.68 without) [src: DER§3.4].

PLACEHOLDER: G1 result on seeds 2/3 (both pairs, matched sc-inr / NoSinc).

Predicting the loss from the full model (L2, pre-registered). One account is that the cell or footprint input tells the decoder the scale factor, so a model without it behaves like the full model averaged over the training scales. We test this without training: average the full model's output over cells corresponding to $s' \in \{1.0, 1.1, \ldots, 4.0\}$ and predict the ablation loss as $L_{\mathrm{pred}} = R_{\mathrm{full}} - R_{\mathrm{marg}}$ on the 27 F1 cells, seed 1 [src: L1L2]. The ranking matches: Spearman$(L_{\mathrm{pred}}, L) = 0.822$ (LTE) and 0.824 (SC-INR), against 0.40–0.49 for the zone predictors of F1 and $+0.147/+0.149$ above the footprint gain ϵ_b^2 (0.676 on this evaluation) [src: L1L2]; with log-uniform s' it is 0.805/0.803. Two pre-registered criteria fail. The magnitude matches only at $\times 4$ and above: the median L_{pred}/L is 0.59 (LTE) and 0.36 (SC-INR), outside $[0.5, 2]$ for SC-INR, because at $\times 2/\times 3$ the ablation loses more than marginalisation predicts (e.g. BSD100 $\times 3$ b_3: 0.038 vs. 0.008). The ablated output is closer to the marginalised than to the full output in 73.9% / 69.4% of image–scale pairs (threshold 80%), rising with scale (LTE: 58% at $\times 2$, 99% at $\times 8$) [src: L1L2]. So the ablation loss can be predicted after the fact from the full model's cell average, consistent with the readout supplying scale information at large scale factors; at small scale factors the ablation loses more than that, and no per-frequency linear account reproduces it [src: DER§3.4]. This is correlational evidence; a causal test needs a pure scalar-s embedding arm, which requires training and has not been run.

Reading. The footprint readout pays off where the network has recovered content, not where a linear estimator could: in the linear regime C, the learned prior lifts these bands to intermediate recovery, and only there does the readout matter. Of the explanations we tested, scale marginalisation (L2) ranks the losses best; the zones supply only the boundary condition (no loss at $R_b^{\mathrm{net}} \approx 0$ or 1). The three regimes describe the inside of an ASISR network only when assigned by R_b^{net}, which is post-hoc (§4). We do not claim that R_b^{lin} predicts this component.

5.6 Summary of predicted vs. observed winners (E4)

PLACEHOLDER: Table of predicted vs. observed winners across all settings (E4-T2). Ready to transcribe except the E1 rows (pending the corrected prediction) and the F1 rows (seed 1 corrected; seeds 2/3 pending G1). Current tally from E4: one clear mismatch, multi-frame $\sigma_n = 0.005$.

6 Limitations

- **Theory is classical.** Props. 2 and 3 are direct consequences of Wiener/LMMSE theory. Our contribution is their use as an a-priori diagnostic and prescription for spectral adaptation in continuous representations, with cross-task tests.
- **Linear bounds do not bound learned priors.** Trained ASISR networks exceed the oracle linear bound in every regime-C cell (§5.2); we report this as a finding.
- **The a-priori budget does not predict components inside a network.** F1's pre-registered predictor failed (§5.5); assigning regimes by R_b^{net} needs ground truth, and at the network layer Prop. 3 gives only a boundary statement.
- **Threshold sensitivity.** "Decoders have no leverage in regime C" holds at a range threshold of 0.03 (10/12 cells) but not at 0.02 (8/12: $\times 2$ b_3 Set14 0.038, Urban100 0.035; $\times 8$ b_2 Set14 0.021, Urban100 0.026) [src: E4-E3sens].

  PLACEHOLDER: sensitivity of the regime thresholds 0.1/0.8.
- **Multi-frame switch point.** The switch from low-bias to spectral-prior arms occurs at $\sigma_n = 0.005$, earlier than pre-registered (P1 fails), and the generic prior does not reach within 0.6 dB of the oracle at high noise (P2 fails) [src: E1]. The per-frequency diagnostic (D11) supports the band-wise account for GP arms (rank correlation 0.856) but the predicted cost does not select the best arm (1/3), and the nonlinear arms were not diagnosed; zone labels alone should not be used to pick a winner.

- **Scalar model.** Random sparse sampling spreads aliasing across bands; the one asymmetry counter-example (sparse 10% b_1) lies there.
- **Scale collapse and readout-as-scale.** L1's main criterion fails on Set14 (6/9 groups within 0.05; cause undiagnosed). L2 ranks ablation losses well, but its magnitudes miss at $\times 2/\times 3$, and it is correlational: no scalar-s arm has been trained [src: L1L2].
- **Seeds.** F1 currently uses one seed; seeds 2/3 are tested under the frozen G1 pre-registration.

  PLACEHOLDER: update when seeds 2/3 finish.
- **Known operator.** A must be known; blind degradation is out of scope. The pre-registration did not fix the band-energy window in E2 (Hann window added after a one-image run; §5.1).

7 Conclusion

PLACEHOLDER: Write after E1 explanation and G1 and D11 results are in. Intended message: decide where to spend effort by computing R_b^{lin} first; rank representations by the band-summed cost $C(Q)$ with per-band κ_b, not by zone labels; in C only learned priors help; ASISR decoder design sits in C, which explains its saturation, and SC-INR's readout acts in the bands the learned prior lifts out of C.

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A Proof of Proposition 3

With $R = p/(p+q)$ the matched gain R attains error $p(1-R)$. The gain $g = \kappa p/(\kappa p + q) = \kappa R/(\kappa R + 1 - R)$ gives error $(1-g)^2 p + g^2 q$. Using $q/p = (1-R)/R$ and $D = \kappa R + 1 - R$,

$$\frac{(1-g)^2 p + g^2 q}{p} = \frac{(1-R)\big[(1-R) + \kappa^2 R\big]}{D^2}.$$

Subtracting $(1-R)$ and expanding D^2, the numerator becomes $(1-R)\,R(1-R)(\kappa-1)^2$, which is Eq. (2). For Cor. 1, $\Delta(R,\kappa)/\Delta(R,1/\kappa) = \big((1-R)\kappa + R\big)^2 / \big(1 - R + \kappa R\big)^2$, which exceeds 1 for $\kappa > 1$ iff $(\kappa-1)(1-2R) > 0$, i.e. iff $R < 1/2$ (also checked numerically).

B Reproducibility

PLACEHOLDER: Code paths, pre-registration hashes and per-experiment JSON files; see SOURCES.md in the draft folder for the current mapping.